Sixian Shen

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Research Interests

- **AI for Science** Protein Language Models Cryo-EM
- **Computer Vision** 3D Vision Scene Generation
- **Generative Modeling** Diffusion Models Probabilistic Graphical Models

Education

Exchange Student in Computer Science Aug. 2026–Dec. 2026
University of California, Berkeley

B.Sc. in Computer Science 2024–Present
GPA: 3.77/4.00 ShanghaiTech University

Experience

Research Intern StepFun, Co., Ltd. Jun. 2025–Sep. 2025
Research Intern VRVC Lab, supervised by Prof. Jingyi Yu Sep. 2025–Present

Projects

- **ProtLingo: Towards Efficient Protein Language Models.** Jun. 2026–Present
  Conducted a comprehensive analysis and systematic improvement upon cutting-edge models (ESMs). Co-led the development of **ProtLingo**, an efficient protein language modeling framework for sequence modeling. Key contributions include the design of a centered latent N-gram memory module for reusable local-context retrieval, the integration of sparse Mixture-of-Experts upcycling for residue-dependent computation, and the establishment of a parameter-efficient continuation-training pipeline based on pretrained ESM2 backbones. Conducted comprehensive experiments across protein fitness prediction, FLIP benchmarks, supervised contact prediction, and biological specialization analyses, demonstrating that ProtLingo achieves competitive performance with a compact active-parameter budget while preserving structure-relevant representations and is available as an arXiv preprint. Ongoing efforts focus on training ProtLingo from scratch at larger scale, aiming to democratize efficient protein language modeling for downstream biological tasks while improving representation quality, scaling behavior, and task generalization.

- **CryoACE & CryoAtlas: Towards Cryo-EM Foundation Models** Sep. 2025–Feb. 2026
  Cryo-electron microscopy (cryo-EM), recognized by the 2017 Nobel Prize in Chemistry, has become a central technology for resolving protein structures from experimental density maps. Built a large-scale cryo-EM dataset by curating over 20,000 paired density maps and atomic structures from National Facility for Protein Science in Shanghai, Protein Data Bank (PDB) and Electron Microscopy Data Bank (EMDB). Designed and implemented preprocessing pipelines for sequence–structure alignment, structure–density registration, Q-score computation, map denoising, and benchmark construction, resulting in one of the largest cryo-EM datasets and evaluation suites to date. Contributed to training and improving ModelAngelo on this dataset, achieving enhanced automated model-building performance; this work contributed to **CryoAtlas**, accepted to the *NeurIPS 2026 Evaluations & Datasets Track*. Contributed to developing **CryoACE**, a neural pipeline for automated atomic model building from Cryo-EM maps by integrating structural priors from AlphaFold3 with heterogeneity modeling from cryo-EM foundation models. CryoACE resolves continuous conformational states directly from experimental density, enabling dynamic mechanistic interpretation beyond static structures, and was accepted to *ICML 2026*.

- **SPREAD: Physically Plausible 3D Scene Generation** Aug. 2025–Mar. 2026
  Contributed to SPREAD, a diffusion-based 3D scene generation framework that jointly models spatial relations, physical interactions, and geometry-aware object layouts for embodied AI environments. Responsible for the **multi-guidance framework**, implementing differentiable collision, gravity, and relation guidance to reduce interpenetration, floating artifacts, and relational inconsistency during sampling. Led systematic evaluation on 3D-FRONT and ProcTHOR, covering FID, mesh collision rate, graph recall, average support distance, and NVIDIA Isaac Sim stability. Also participated in dataset preprocessing and relation annotation, including scene decomposition, object filtering, mesh correction, grounding, and support/spatial-relation extraction. The method was accepted to *CVPR 2026*.

Publications

1. ProtLingo: Efficient Protein Language Modeling via Conditional Memory and Expert Routing.
   Mingrui Li*, **Sixian Shen***, Minzhang Li, Ruiyi Zhang, Kexin Zhang, Jiakai Zhang, Jingyi Yu.
   [*arXiv preprint, 2026* | arXiv]
2. SPREAD: Spatial-Physical REasoning via geometry Aware Diffusion.
   Minzhang Li*, Kuixiang Shao*, Xuebing Li, Yuyang Jiao, Yinuo Bai, Hengan Zhou, **Sixian Shen**, Jiayuan Gu, Jingyi Yu.
   [*CVPR, 2026* | Code | Paper]
3. CryoACE: An Atom-centric Framework for Accurate and Automated Model Building in Cryo-EM.
   Minzhang Li, Mingrui Li, Weichen Qin, Qihe Chen, **Sixian Shen**, Yuan Pei, Jiakai Zhang, and Jingyi Yu.
   [*ICML, 2026* | Code | Paper]
4. CryoAtlas: A Large Curated Dataset and Unified Benchmark for Cryo-EM Atomic Model Building.
   Mingrui Li, Minzhang Li, Weichen Qin, Yufan Xie, Zhangzhi Xiong, **Sixian Shen**, Jiakai Zhang, Yuan Pei, Jingyi Yu.
   [*NeurIPS Evaluations & Datasets Track, 2026* | Project | Dataset]
5. Evolution-Aware MSA Reasoning for Subsampling via Factor Graphs.
   Zhangzhi Xiong*, Minzhang Li*, Haotian Yu*, **Sixian Shen**, Kexin Zhang, Mingrui Li, Jie Zheng, Kewei Tu, Jingyi Yu.
   [*arXiv preprint, 2026* | arXiv]

Awards

Honorable Mention, Interdisciplinary Contest in Modeling (ICM) 2026
Shanghai Municipal Second Prize, China Undergraduate Mathematical Contest in Modeling (CUMCM) 2025

References

Prof. Jingyi Yu	Supervisor, IEEE Fellow	ShanghaiTech University	yujingyi@shanghaitech.edu.cn
Minzhang Li	Student Mentor, Ph.D.	ShanghaiTech University	limzh2022@shanghaitech.edu.cn